

GitHub Explorer Project

GitHub Explorer is an interactive dashboard built with React.js, allowing users to explore trending GitHub repositories with analytics. It includes features such as search, sorting, filtering, bookmarking, and note-taking, along with visualizations using Chart.js. The interface is fully styled using Tailwind CSS for a responsive design.

Features:

- Search repositories by name
- Sort by Stars, Forks, or Last Updated
- Filter by language (JavaScript, Python, Java, TypeScript, C++, etc.)
- Repository Cards displaying repository name, description, stars, forks, and language
- Bookmarking & Notes saved in localStorage
- Analytics: Bar chart showing Top 5 repositories by stars using Chart.js
- Responsive UI styled with Tailwind CSS

Tools & Technologies:

- React.js – Frontend framework
- GitHub REST API – Fetch trending repositories
- Chart.js + react-chartjs-2 – For analytics visualizations
- Tailwind CSS – Styling framework
- LocalStorage – Saving bookmarks and notes

Installation & Setup:

1. Clone the repository

```
git clone https://github.com/YOUR-USERNAME/github-explorer.git  
cd github-explorer
```

2. Install dependencies

```
npm install
```

3. Run the development server

```
npm run dev
```

4. Open in browser

<http://localhost:5173>

Project Structure:

github-explorer/

```
├─ src/
│   ├── App.jsx      # Main React component
│   ├── main.jsx     # Entry point
│   ├── index.css    # Tailwind CSS import
│   └─ assets/       # Optional folder for screenshots or images
├─ package.json      # Node dependencies
├─ tailwind.config.js # Tailwind configuration
├─ postcss.config.js  # PostCSS configuration
└─ README.md         # Project documentation
```

Screenshots:

- Dashboard with search, sorting, and filtering
- Repository Cards
- Analytics Chart
- Bookmarks & Notes Section

Usage:

- Search: Type in the search bar to filter repository names.
- Sort: Select sorting options from the dropdown (Stars, Forks, Updated).
- Filter by language: Choose a programming language from the dropdown.
- Bookmark/Notes: Click the “Bookmark” button to save a repo, add notes in the textarea.
- Analytics: Top 5 repositories by stars are displayed in a bar chart at the bottom.

Deployment (Optional):

1. Go to Vercel and sign in with GitHub.
2. Click New Project → import your github-explorer repository.
3. Keep default settings (Vite + React detected automatically).
4. Click Deploy.
5. After deployment, you will get a live URL like: <https://github-explorer.vercel.app>

Conclusion:

GitHub Explorer demonstrates a full-featured React project that interacts with a public API, includes data visualization, state management, bookmarking, and notes functionality. The responsive interface with Tailwind CSS ensures a professional look on both desktop and mobile devices.

License:

This project is free to use and modify.